

Math Olympiad Trainer Academy

AY 25/26 — Cycle IV | Exercise Collection

Session 1 — Olympiad Number Theory

Exercise 1

Find all primes p such that

$$\frac{2^{p-1}-1}{p}$$

is a perfect square.

Exercise 2

Find all positive integers a, b such that

$$3^a - 2^b = 1.$$

Exercise 3

Find all positive integers a, b such that

$$3^a + 2 = 5^b.$$

Exercise 4

Find $\text{ord}_{1001}(2)$.

Exercise 5

Find all positive integers n such that

$$n \mid 2^n - 1.$$

Exercise 6

Let $n \in \mathbb{N}$. If

$$n \mid 2^n + 1,$$

prove that

$$n = 1 \text{ or } 3 \mid n.$$

Exercise 7

When does

$$2^a - 1 \mid 2^b - 1$$

hold?

Exercise 8

When does

$$2^a + 1 \mid 2^b + 1$$

hold?

Exercise 9

Let a be an integer and p a prime. Suppose a prime q divides

$$S = a^{p-1} + a^{p-2} + \dots + a + 1.$$

Prove that

$$q \equiv 1 \pmod{p} \text{ or } q = p.$$

Exercise 10

Find all positive integers n such that all three conditions hold:

$$n \mid 2^n, n-1 \mid 2^n - 1, n-2 \mid 2^n - 2.$$

Exercise 11

Let $a, b \in \mathbb{Z}$, and let p be a prime. If

$$a \equiv b \pmod{p},$$

prove that

$$a^p \equiv b^p \pmod{p^2}.$$

Exercise 12

Define

$$a_n = 2^n + 3^n + 6^n - 1, n = 0, 1, 2, \dots$$

Find all primes that do not divide any term of this sequence.

Session 2 — Valuations and Lifting the Exponent

Exercise 13

What is the largest n such that

$$5^n \mid 3^{1000} - 2^{1000}?$$

Exercise 14

What is the largest power of 2 that divides

$$7^{80} - 1?$$

Exercise 15

Evaluate

$$v_3(2^{3^{10}} + 1).$$

Exercise 16

If $p \mid a - b$, but $p \nmid a, b$, what is

$$v_p\left(\frac{a^p - b^p}{a - b}\right)?$$

Exercise 17

If $p \nmid a - b$, $p \nmid a, b$, and $p \nmid n$, what is

$$v_p\left(\frac{a^n - b^n}{a - b}\right)?$$

Exercise 18

What is

$$v_{11}(3^{100!} - 1)?$$

Exercise 19

What is the order of 2 modulo 3^{100} ?

Exercise 20

Prove that there exists $n \in \mathbb{N}$ such that

$$2^n \equiv 5 \pmod{3^{100}}.$$

Exercise 21

Find all primes p and $m, n \in \mathbb{N}$ such that

$$(p-1)^m + 1 = p^n.$$

Exercise 22

Find all $a, b \in \mathbb{N}$ for which

$$7^a = 3 \cdot 2^b + 1.$$

Exercise 23

Find all $n \in \mathbb{N}$ for which

$$n^2 \mid 2^n + 1.$$

Exercise 24

Find all pairs of odd $a, b \in \mathbb{N}$, $a, b < 2^{100}$, such that $a^b + b$ and $b^a + a$ are divisible by 2^{100} .

Exercise 25

Find all $x, y, n \in \mathbb{N}$ and p prime such that

$$x^p + y^p = p^n.$$

Exercise 26

Let $n \in \mathbb{N}$ be given. Find the greatest common divisor of all numbers of the form

$$x^n + (x+1)^n + (x+2)^n,$$

where $x = 1, 2, 3, \dots$

Session 3 — Primitive Roots and Applications

Exercise 27

Is there a primitive root modulo 143?

Exercise 28

We say that n cells in an $n \times n$ table are “scattered” if they lie in different rows and different columns. Is it possible to write $1, 2, 3, \dots, n^2$ in the table so that the product in any n scattered cells modulo $n^2 + 1$ is always the same, if:

1. $n = 8$?
2. $n = 10$?

Exercise 29

Find all $n \in \mathbb{N}$ for which

$$(n-1)! + 1 = n^k, k \in \mathbb{N}.$$

Exercise 30

Prove that there is an odd $a \in \mathbb{N}$ such that

$$2^n + a$$

is composite for all $n = 1, 2, 3, \dots$

Exercise 31

What is the smallest sum of digits of a positive multiple of 3^{100} ?

Exercise 32

Find all pairs of primes p, q such that

$$a^{3pq} \equiv a \pmod{3pq}$$

holds for every integer a coprime to $3pq$.

Exercise 33

Find all $n \in \mathbb{N}$ such that

$$n \mid 2^{n-1} + 1.$$

Exercise 34

Find all $a \in \mathbb{N}$ for which

$$1, 1+a, 1+a+a^2, 1+a+a^2+a^3, \dots$$

take all possible remainders modulo $10!$.

Exercise 35

Find all $x, y \in \mathbb{N}$ such that

$$3^x - 8^y = 2xy + 1.$$